

Sales Data Analysis and Forecasting

Professional Project Report

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Original Records 51,298	Cleaned Records 51,290	Countries 165	Final R² 0.598
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1. Project Overview

This project analyzes a global sales dataset to identify sales trends, regional performance, product performance, profitability patterns and future sales trends. The workflow covers data cleaning, exploratory data analysis, visualization, correlation analysis and machine learning-based sales forecasting.

2. Objectives

- Clean and prepare the raw sales dataset
- Handle missing values and incorrect data types
- Perform exploratory data analysis
- Analyze sales by category and region
- Analyze monthly sales and profit trends
- Identify top-performing products
- Study relationships between sales, profit, discount and shipping cost
- Build and evaluate sales forecasting models
- Present business-friendly insights

3. Dataset and Data Preparation

The original dataset contained 51,298 records and 21 columns. After removing 8 invalid rows the cleaned dataset contained 51,290 records. No duplicate rows were identified. The dataset includes order, customer, product, sales, quantity, discount, profit, shipping cost and order priority information.

Order Date and Ship Date were converted to datetime format. Missing postal codes remain in the cleaned data because postal codes were not required for the main sales analysis. The cleaned dataset was saved as cleaned_sales.csv.

4. Exploratory Data Analysis

- **Category performance:** Technology generated the highest total sales among the three major categories followed by Furniture and Office Supplies.
- **Regional distribution:** The dataset contains 165 countries across America, Europe, Asia, Africa and Oceania. America has the highest number of records.
- **Product performance:** Technology products particularly smartphones appear prominently among the top-performing products by total sales. Apple Smart Phone, Full Size was the highest-selling product among the top 10 products analyzed.
- **Sales and Profit:** The correlation between Sales and Profit is approximately 0.485 which indicates a moderate positive relationship.

- **Discount and Profit:** The correlation is approximately -0.316 which indicates that higher discount levels are generally associated with lower profitability.
- **Shipping Cost and Sales:** The correlation is approximately 0.77 which indicates a strong positive relationship between the two variables.

5. Sales Forecasting

Monthly sales were aggregated to create a time-based forecasting dataset. Linear Regression and Random Forest Regression were evaluated. An improved Linear Regression model was then developed using lag features representing previous sales values.

6. Model Comparison

Model	MAE	RMSE	R ²
Linear Regression	51,799.47	62,484.63	0.153
Random Forest	78,192.13	90,176.88	-0.763
Linear Regression with Lag Features	36,656.75	43,625.22	0.598

7. Final Model

The Linear Regression model with lag features was selected as the final model because it achieved the best performance among the tested models. It used Lag_1 for previous month's sales, Lag_2 for sales from two months earlier and Lag_12 for sales from the same month in the previous year.

- MAE: 36,656.75
- RMSE: 43,625.22
- R²: 0.598

The final model explains approximately 59.8% of the variation in the test-period monthly sales.

8. Business Insights

- Technology is the strongest major category by total sales.
- America has the largest number of sales records in the dataset.
- Technology products particularly smartphones are important contributors to sales.
- Higher sales generally correspond with higher profit but sales alone do not fully determine profitability.
- Higher discounts are generally associated with lower profitability.
- The strong relationship between Sales and Shipping Cost suggests that larger transactions tend to have higher shipping costs.
- Historical sales information improved forecasting performance compared with the baseline models tested.

9. Conclusion

This project demonstrates a complete sales analytics and forecasting workflow from raw data preparation through exploratory analysis and machine learning. The analysis identified important patterns in category performance, regional distribution, profitability, discounts and product sales. Among the tested models the Linear Regression model with lag features provided the best performance and was selected as the final forecasting model.

10. Project Deliverables

- Cleaned sales dataset
- Jupyter Notebook containing the analysis and forecasting workflow
- README.md with project documentation
- Visual analysis and model evaluation
- Final forecasting model results
- Professional project report

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Note: Detailed charts and code remain available in the project Jupyter Notebook. This PDF provides the consolidated written report and final analytical results.